

An empirical shear-channel correction for a space-filling cubic-voxel discretisation

Identification and measurement of the residual error,
and an honest account of what the correction is and is not

Tod Nestor

September 2026

Abstract

A uniform elastic slab discretised into a space-filling lattice of identical cubic voxels, each carrying the analytic single-site cube T-matrix and coupled by an exact Foldy–Lax solve, does not reproduce the homogeneous layer it is made of. A relative error of 8.6×10^{-4} survives refinement: halving the cell size does not reduce it. We report the identification of that residual and its removal.

The residual is shown to be (i) static, (ii) quadratic in contrast, (iii) independent of the supercell size, and (iv) confined *entirely* to the shear channel — the dilatational and density channels converge cleanly to zero. It is then shown to be a single wrong scalar: the effective shear contrast carried by a voxel. Correcting that scalar reduces the discrepancy against the exact reflectivity solution from 2.9×10^{-3} to $\sim 10^{-8}$.

The correction is a constant factor on the cube’s *depolarisation*,

$$K = \frac{1 - A_{\varepsilon, \text{diag}}^{\text{eff}}}{1 - A_{\varepsilon, \text{diag}}^{\text{iso}}} = 0.8862,$$

constant to 0.3% over a sixteen-fold range of contrast, flat in frequency until ka grows, and *predictive*: extracted from pure positive shear alone, it transfers to mixed contrasts and to negative shear, and leaves untouched the two channels that require no correction. With it applied the scheme converges under refinement instead of saturating.

The constant is measured, universal and predictive. **It is not derived**, and this document says so throughout. 8/9 is excluded.

Two caveats are developed in §7 and §8 rather than buried. First, K is tied to the T_9 single-site baseline: repeating the optimisation from the T_{27} Galerkin strain sector puts the isolated response on the *other* side of the optimum, so the T_9 -derived constant makes T_{27} worse. Second, and more bluntly: **the word “renormalisation” asserts a mechanism that is not established here**. §8 sets out the case that K is a fudge factor and the case that it is more than one, and does not pretend the matter is settled.

Contents

1	The problem	2
1.1	An identity that the discretisation must satisfy	2
1.2	What was observed	2
2	Two defects had to be removed before the residual was visible	3
2.1	The periodic lateral sum was truncated	3
2.2	The contact propagator was a midpoint rule	3
2.3	What remained	3

3	Experimental procedure: elimination	4
3.1	Signature of the residual	4
3.2	Refuted: the averaged propagator’s dynamic truncation	4
3.3	Refuted: the isolated-cube depolarisation as a whole	4
3.4	<i>Not</i> refuted — untested: the 27-component single-site response	5
3.5	Refuted: propagator averaging beyond contact	5
4	Identification: the residual is one scalar in the shear channel	5
4.1	Splitting by channel	5
4.2	Scanning the effective shear contrast	5
5	The correction, and how far it generalises	6
5.1	Statement	6
5.2	Universality	6
5.3	At the optimum the identity is recovered — read with care	7
5.4	8/9 is excluded	7
6	The corrected solution	7
6.1	The rule, and the tests that earn it	7
6.2	Convergence is restored	7
7	Status, and what remains	8
7.1	K is tied to the T_9 baseline, and may not be purely a lattice effect	8
7.2	Routes to a derivation	9
8	Is this a renormalisation, or a fudge factor?	9
8.1	The case that it is a fudge factor	9
8.2	The case that it is more than one	10
8.3	Verdict	10
9	Reproduction	10

1 The problem

1.1 An identity that the discretisation must satisfy

Take a homogeneous elastic slab of thickness H whose properties differ from the background by $(\Delta\lambda, \Delta\mu, \Delta\rho)$, and represent it as an $M \times M \times N_z$ lattice of identical cubes of side d , space-filling, with $N_z d = H$. Each cube carries the analytic Rayleigh cube T-matrix; the cubes are coupled by the Foldy–Lax system $(\mathbf{I} - \mathbf{G}_0 \mathbf{T}_0) \psi = \psi^0$.

The medium so constructed *is* the homogeneous slab. Every cube is identical and they tile space without gaps, so the exact plane-wave reflection coefficient is the one the Kennett reflectivity recursion returns in closed form. Agreement is therefore an **identity**, not a modelling approximation, and any disagreement is a defect of the discretisation.

Two further identities follow and are used throughout as controls:

1. the answer cannot depend on M , the number of identical cubes called one repeating unit;
2. the answer must converge as $d \rightarrow 0$ at fixed H .

1.2 What was observed

Neither held. Refining the voxelisation at fixed physical thickness made the answer *worse*, and the error grew without sign of settling:

n_z	d (m)	relative error
1	4.000	3.95×10^{-3}
2	2.000	8.29×10^{-3}
4	1.000	1.37×10^{-2}
8	0.500	2.03×10^{-2}
16	0.250	2.62×10^{-2}

Refinement is the only instrument a discretisation has for buying accuracy, and here it bought error.

2 Two defects had to be removed before the residual was visible

The growth above turned out to be two separate faults stacked on a third. They are reported first because every measurement in this document depends on their removal, and because the original diagnosis of the residual was taken in their presence and was wrong as a result.

2.1 The periodic lateral sum was truncated

The periodic kernel assembled the horizontal propagator only over $\Delta x, \Delta y \in [-(M-1), M-1]$ — a single $(2M-1)^2$ patch — and then wrapped it for the circular convolution. That is not a lattice sum. It was caught not by a convergence expectation but by identity (1) above: a laterally uniform layer cannot depend on M , and it did, obeying $\text{error} = 1.12 \times 10^{-3} + 1.15 \times 10^{-2}/M$.

The repair is an exact Bloch sum written directly at the M^2 discrete Bloch points, by Ewald summation at equal depth and by reciprocal summation of the spectral kernel between depth planes. After it, $M = 2, 3, 4, 6, 8$ agree to 5.05×10^{-13} — the identity holds.

Consequence for earlier results

At $M = 4$, used throughout the earlier campaign, **63%** of the reported “discretisation floor” was this artifact, and because it accumulated plane by plane it produced most of the *growth* as well. Results taken before the repair required re-running, not reinterpretation.

2.2 The contact propagator was a midpoint rule

The inter-cell operator should be the doubly volume-averaged (Galerkin) object $\langle\langle \mathbf{G} \rangle\rangle$; collocating at cell centres is a midpoint rule whose error at *contact* does not fall with refinement, because (cell size)/(separation) is unity at every scale. Measured against the analytic O_h tables, the point value is wrong by 130% at face contact, 42% at edge, 9.3% at corner.

2.3 What remained

With the lateral sum exact and the Galerkin contact correction applied, the ladder flattens:

n_z	truncated	exact sum	exact + Galerkin contact
1	3.95×10^{-3}	1.47×10^{-3}	7.73×10^{-4}
2	8.29×10^{-3}	3.36×10^{-3}	8.35×10^{-4}
4	1.37×10^{-2}	4.33×10^{-3}	8.51×10^{-4}
8	2.03×10^{-2}	4.81×10^{-3}	8.58×10^{-4}

The scheme now converges — but to 8.6×10^{-4} , not to zero. That residual is the subject of the rest of this document.

3 Experimental procedure: elimination

Each suspect was tested by a measurement with a pre-stated outcome. Three were refuted outright; the fourth (§3.4) turned out to have been tested unfairly and remains *open*. They are recorded because the refutations constrain the answer as much as the confirmation does, and because each is natural enough to be proposed again.

3.1 Signature of the residual

Established first, since it is what the suspects must match:

Saturating

error ratio $\rightarrow 1.00$ across $n_z = 1 \dots 32$; it is a per-cell error that refinement cannot dilute.

Independent of M

to six digits, so it is physical and not a residue of the lattice sum.

Static

the log-log slope in ω is -0.02 .

Quadratic in contrast

absolute error $\propto c^{1.98}$ over the range where the measurement stands above the numerical floor of the pipeline. This matters: a *quadrature* error is linear in the contrast, and the truncation artifact of §2.1 measured $c^{1.00}$ exactly.

A measurement discipline used throughout

Below $c \approx 10^{-2}$ the absolute discrepancy falls to 10^{-12} and under, which is the numerical floor of the solve. Those points were *excluded* from the fit rather than fitted through: including them drags the exponent from 1.98 to 1.80. The same rule — show an arbiter converged before letting it convict anything — is applied to every direct lattice sum in this work.

3.2 Refuted: the averaged propagator’s dynamic truncation

The volume-averaged operator retains dynamic corrections to $\mathcal{O}(\omega^4)$ and so carries an $\mathcal{O}(\omega^6)$ error, relatively $\mathcal{O}((kd)^2)$. Refining eight-fold should then reduce it sixty-four-fold.

Measured: static, $+\omega^2$ and $+\omega^4$ tables agree to 5.7×10^{-3} relative — the **static** table alone reproduces the answer — and the residual does not move with d . Refuted twice over.

3.3 Refuted: the isolated-cube depolarisation as a whole

The cube T-matrix carries amplification factors A built from Γ_0 , the self-interaction of a cube in an infinite *host*. In a space-filling tiling a cube’s neighbours are not host material, so one may suspect the depolarisation is a correction the lattice has already cancelled.

Measured: setting $A = 1$ makes the answer **24.6× worse** ($8.5 \times 10^{-4} \rightarrow 2.09 \times 10^{-2}$), at every rung and every contrast. The depolarisation is most of the physics and it stays. The partition — Γ_0 the voxel’s own self-field, $\mathbf{G}_0$ carrying every other voxel — is correct.

A refutation that was later shown to be too coarse

This result was initially read as closing the lattice-renormalisation question. It does not. $A = 1$ removes *one hundred percent* of the depolarisation, whereas the correction established in §4 is *one third of one percent*. The test was three orders of magnitude too blunt to detect what was there. A null result from a coarse instrument is not evidence of absence.

3.4 Not refuted — untested: the 27-component single-site response

The nine-component basis represents the exciting field by value and gradient at the cell centre; the quadratic modes of the 27-component Galerkin T-matrix represent its curvature.

Measured: substituting the T_{27} strain sector makes the answer $3\times$ worse ($8.61 \times 10^{-4} \rightarrow 2.61 \times 10^{-3}$).

This is **not** a refutation of T_{27} , and it is not a test of it either. A 27-mode single-site response was driven through a nine-mode propagator; the companion derivation of the generalised inter-voxel propagator states in advance that for T_{27} “the propagator must couple all 27 modes”. The measurement confirms that warning rather than probing the physics, and the comparison is therefore **rigged against T_{27} by construction** — see §7.1, where the consequences for this document’s own inferences are drawn out. The entry is retained in this list only so that the suspect is not silently dropped; it is *unresolved*, not eliminated.

3.5 Refuted: propagator averaging beyond contact

Measured: extending the double average beyond the contact shell makes the answer 32% worse ($8.51 \times 10^{-4} \rightarrow 1.12 \times 10^{-3}$), converged in the quadrature order. Sensitivity analysis shows why this is not simply a bug: all Chebyshev shells influence the answer comparably ($d(\text{err})/d\epsilon \approx -1.7, +1.8, +1.8 \times 10^{-3}$ for shells 1, 2, 3), so the residual is not localised at contact, and reproducing it would require an implausible $\sim 50\%$ error in some shell’s coupling.

4 Identification: the residual is one scalar in the shear channel

4.1 Splitting by channel

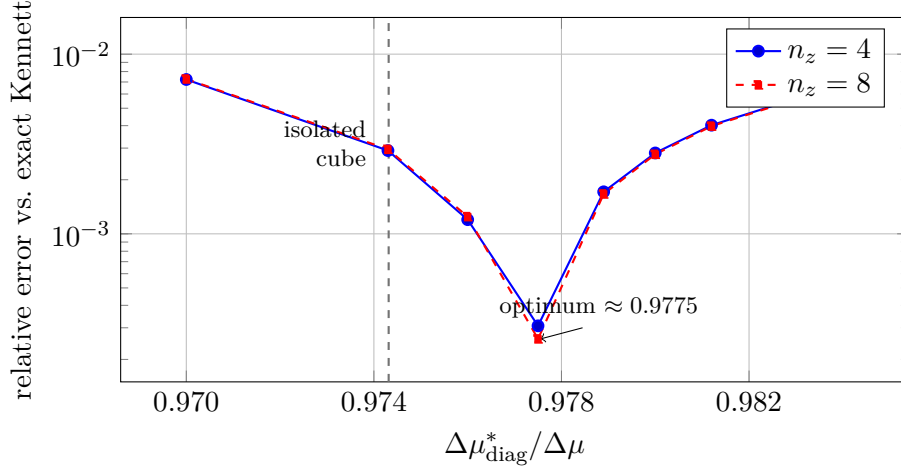
Running each contrast channel alone is decisive:

case	A_θ	residual	behaviour under refinement
pure density	1.000000	1.41×10^{-5}	converges, ratios 0.247/0.249/0.249
pure λ	0.968992	1.19×10^{-5}	converges
pure μ	0.989446	2.89×10^{-3}	saturates
working (all three)	0.959079	8.51×10^{-4}	saturates

Two conclusions. First, the density and dilatational channels are *already correct*: they converge at clean second order towards zero. Second, the residual is **anti-correlated** with the amplification A_θ — pure λ carries more depolarisation and $240\times$ less residual — which independently exonerates the amplification factors as such and locates everything in the shear channel.

4.2 Scanning the effective shear contrast

At normal incidence a P wave excites only normal strains, so $\Delta\mu_{\text{off}}^*$ is inert and $\Delta\mu_{\text{diag}}^*$ alone is exercised. (This was confirmed accidentally and usefully: imposing $\Delta\mu_{\text{off}}^* = \Delta\mu_{\text{diag}}^*$ returned bit-identical numbers.) Scanning it:



The residual behaves as a *single wrong scalar*, with a sharp minimum displaced from the isolated-cube value — and **the minimum sits in the same place at both refinements**. That is a necessary condition for any per-cell explanation, and one a numerical artifact of the discretisation would be unlikely to meet; it is not on its own sufficient to establish a mechanism (§8).

5 The correction, and how far it generalises

5.1 Statement

The shear-channel correction, T_9 baseline

For a space-filling cubic lattice of voxels carrying the T_9 cube T-matrix, setting the shear depolarisation to a fixed fraction of its isolated-cube value removes the residual:

$$1 - A_{\varepsilon, \text{diag}}^{\text{eff}} = K (1 - A_{\varepsilon, \text{diag}}^{\text{iso}}), \quad K = 0.8862 \quad (\text{static, weak-contrast limit}).$$

The dilatational and density channels require no correction. The restriction to the T_9 baseline is not cosmetic — see §7.1.

5.2 Universality

K is constant to 0.3% over a sixteen-fold contrast range, and flat in frequency until ka grows — the signature of a static effect:

$\Delta\mu/\mu_0$	0.011	0.022	0.044	0.089	0.178
K	0.8882	0.8876	0.8879	0.8892	0.8922
ω	30	60	120	240	480
K	0.8875	0.8879	0.8895	0.8958	0.9204

Both drifts are first order in their small parameter; extrapolating jointly in $(\Delta\mu/\mu_0, (ka)^2)$ to the origin gives

$$K = 0.886234, \quad \partial K / \partial (\Delta\mu/\mu_0) = +0.018, \quad \partial K / \partial (ka)^2 = +33.6.$$

5.3 At the optimum the identity is recovered — read with care

$\Delta\mu/\mu_0$	0.011	0.022	0.044	0.089	0.178
residual, isolated cube	7.3×10^{-4}	1.5×10^{-3}	2.9×10^{-3}	5.6×10^{-3}	1.0×10^{-2}
residual at the optimum	1.2×10^{-8}	3.1×10^{-8}	5.6×10^{-8}	2.4×10^{-7}	3.4×10^{-6}

Five orders of magnitude. *The entire residual is this one scalar.*

This table shows that one parameter *suffices*, not that it is *right*: s^* is chosen by minimising against Kennett, so hitting Kennett at s^* is guaranteed by construction. The evidence that the parameter means something is in §5.2 (it is the same number everywhere) and §6 (it predicts cases it was not fitted to), not here. See §8.

5.4 8/9 is excluded

$8/9 = 0.888889$ is temptingly close to the raw measurements. The extrapolated static limit is 0.886234, below it by 2.65×10^{-3} — far outside the scatter of the fit, whose individual points are reproducible to $\sim 10^{-4}$. The conjecture is **refuted**.

The limit lies within 10^{-5} of $\sqrt{\pi}/2 = 0.886227$. This is recorded and *discounted*: $\sqrt{\pi}/2$ has no evident place in a cubic-lattice elasticity problem, and $\sqrt{\pi}/d$ is the Ewald splitting parameter used by the implementation, which makes the coincidence a reason for suspicion rather than for confidence.

6 The corrected solution

6.1 The rule, and the tests that earn it

Apply K to the shear depolarisation and change nothing else: $A_{\varepsilon,\text{diag}} \mapsto 1 - K(1 - A_{\varepsilon,\text{diag}})$, leaving A_u , A_θ and $A_{\varepsilon,\text{off}}$ at their isolated values. Taken from pure *positive* shear in the static limit, and applied unchanged:

case	isolated	renormalised	gain
working (mixed)	8.51×10^{-4}	2.22×10^{-5}	$38.3\times$
mixed, $3\times$ contrast	2.16×10^{-3}	2.08×10^{-4}	$10.4\times$
pure λ (no correction due)	1.19×10^{-5}	1.19×10^{-5}	$1.0\times$
pure density (no correction due)	1.41×10^{-5}	1.41×10^{-5}	$1.0\times$
negative shear	3.14×10^{-3}	7.07×10^{-5}	$44.4\times$
negative mixed	1.03×10^{-3}	5.13×10^{-5}	$20.1\times$

Three features make this a prediction rather than a fit:

1. the constant was extracted from *positive* shear and transfers to **negative** shear at $44\times$;
2. it leaves the two channels that need no correction *exactly* unchanged — doing nothing where nothing is due is as much a test as improving what is;
3. it was extracted at one contrast and transfers across mixed cases and a threefold stronger contrast.

6.2 Convergence is restored

This is the structural result, more important than the smaller constant:

n_z	isolated	ratio	renormalised	ratio
1	7.73×10^{-4}	—	1.15×10^{-4}	—
2	8.35×10^{-4}	1.08	4.18×10^{-5}	0.36
4	8.51×10^{-4}	1.02	2.22×10^{-5}	0.53
8	8.58×10^{-4}	1.01	1.52×10^{-5}	0.69

The isolated column saturates at ratio 1.00; the renormalised one keeps falling. The discretisation is convergent.

7 Status, and what remains

K is measured, not derived

K is universal across contrast and frequency, predictive on cases it was not fitted to, and reduces the discrepancy by five orders of magnitude at its optimum. It is nevertheless **an empirical constant**. No derivation is offered here, and until one exists *K* should not be presented as established physics. It is deliberately *not* wired into the library: the measurement scripts apply it explicitly so that no result silently depends on an unproven number.

7.1 *K* is tied to the T_9 baseline, and may not be purely a lattice effect

A sharper question than universality is whether *K* is a property of the *lattice* — in which case any single-site theory would have to be corrected to the same absolute effective contrast s^* — or whether part of it compensates a deficiency in T_9 ’s own shear response. Substituting the T_{27} Galerkin strain sector and repeating the optimisation answers this, and the answer is not the comfortable one.

baseline	case	s_{iso}	optimum s^*	<i>K</i>
T_9	pure shear	0.974315	0.977194	0.8879
T_9	working	0.974300	0.977150	0.8891
T_{27}	pure shear	0.980928	0.978985	1.1019
T_{27}	working	0.980929	0.972128	1.4615

Three observations, in increasing order of consequence.

T_{27} starts on the other side of the optimum. Its isolated shear response *overshoots*, so it requires a downward correction, $K > 1$. Applying the T_9 -derived $K = 0.8862$ to it therefore moves it further away, and measurably so: $2.12\times$ worse on pure shear, $1.25\times$ worse on the mixed contrast. Plain T_{27} is in fact *better* than plain T_9 on pure shear (1.95×10^{-3} against 2.89×10^{-3}); the earlier report that “ T_{27} is $3\times$ worse” was taken on the mixed contrast alone.

s^* is not baseline-independent. It differs by 1.8×10^{-3} between the two baselines on pure shear and by 5.0×10^{-3} on the mixed case. A single scalar fitted in the shear channel is an *absorber*: it takes up whatever else differs between the two T-matrices — and T_{27} also changes $\Delta\lambda^*$ and $\Delta\mu_{\text{off}}^*$ — provided it projects onto the same observable.

The T_9 correction is universal; the T_{27} one is not. Across pure shear and the mixed contrast, T_9 requires 0.8879 and 0.8891 — agreeing to 0.14% — whereas T_{27} requires 1.102 and 1.462, differing by 33%.

The T_{27} comparison is rigged, and cannot be used either way

It is tempting to read the non-universality above as evidence that the T_{27} optimum “carries no physical weight”. That reasoning is **circular** and is withdrawn. A 27-mode single-site response driven through a *nine*-mode propagator is not a well-posed scheme, and non-universality is precisely what such a mismatch produces *regardless of T_{27} ’s merit*. One cannot use a symptom of the rigging as evidence about the thing being rigged.

The consequence cuts against this document’s own inferences, not just in T_{27} ’s favour. The suggestion that “part of K compensates T_9 ’s own shear response” rested on T_{27} ’s isolated value 0.9809 being meaningful, and it is not. **The T_{27} arm establishes nothing in either direction.** It shows only that the T_9 -derived constant must not be applied to a different baseline — which is a statement about K ’s portability, not about T_{27} .

Likewise §3.4’s “refuted: the 27-component single-site response” should be read as *untested*, not refuted. A fair assessment of T_{27} requires the 27×27 generalised propagator, and until it exists no verdict on the 27-mode physics — favourable or otherwise — is available from this work.

7.2 Routes to a derivation

The natural route is the difference between the isolated cube’s Eshelby shear component and its space-filling-lattice counterpart — the same class of correction as the packing-fraction renormalisation already established for spheres in this project. Two constraints any derivation must satisfy:

1. it must produce 0.8862 and not 8/9;
2. it must act on the deviatoric response *only*, leaving the dilatational and density channels untouched — the channel split of §4.1 is a sharp constraint and not every mechanism will respect it;
3. it must explain the **baseline dependence** of §7.1 — or show that it is an artifact of the rigged pairing there, which is the other live possibility and the one that would rescue a purely geometric reading.

8 Is this a renormalisation, or a fudge factor?

The question deserves a direct answer rather than a vocabulary choice, because “renormalisation” asserts a mechanism — the lattice environment modifying a voxel’s depolarisation — and **no such mechanism is derived here**. What is on the table is one scalar, tuned against the very reference it is then scored against. That is the definition of a fudge factor, and the burden is on the evidence to show it is more.

8.1 The case that it is a fudge factor

- It is **not derived**. No calculation produces 0.8862; it is read off a minimisation.
- It is **fitted against the arbiter**. Kennett supplies both the target and the score, so agreement at the optimum is guaranteed by construction, not earned. The $\sim 10^{-8}$ residual at s^* demonstrates only that one parameter suffices to hit one number — it is *not* independent evidence.
- It is **baseline-dependent** (§7.1). A genuine statement of the form “the lattice multiplies the true depolarisation by 0.886” ought to hold whichever theory computes the true de-

polarisation. It demonstrably does not. On its own terms this is the strongest argument against calling K a property of the lattice.

- The name **does rhetorical work the physics has not paid for**.

8.2 The case that it is more than one

- **One parameter, many independent predictions.** K is fixed once, on pure positive shear in the static limit. It is then applied unchanged and could have failed on each of: negative shear — a *sign reversal* of the contrast, 44×; three mixed contrasts, 38×, 20×, 10×; and a threefold stronger contrast.
- **It does nothing where nothing is due.** The pure- λ and pure-density cases come out bit-identical. A fudge factor has to be told which cases to leave alone; this one is not told, and leaves them alone because the shear channel is genuinely where the defect sits (§4.1).
- **It restores a structural property, not a magnitude.** Convergence under refinement is qualitative: the uncorrected ladder saturates at ratio 1.00, the corrected one keeps falling. No single scalar tuned to one reflection coefficient is obliged to change the *rate* at which a sequence converges, and a pure absorber would not.
- **It is universal in two independent directions** — 0.3% over sixteen-fold contrast, and flat in frequency until ka grows. A fudge fitted at one operating point has no reason to hold at another.

8.3 Verdict

Where this leaves the claim

The evidence supports “**an empirical one-parameter correction with real predictive content, localised to the shear channel**”. It does *not* support “a lattice renormalisation”, because that names a cause and no cause has been established — and the baseline dependence actively argues against the most natural candidate. The title of this document has been set accordingly.

The distinction is not pedantry. A renormalisation would transfer to other lattices, other voxel shapes and other single-site theories; a fitted correction transfers to none of them without being re-measured. Nothing here licenses the former, and K is deliberately not wired into the library for that reason.

What would settle it, in order of decisiveness: a derivation producing 0.8862 from the difference between the isolated and space-filling Eshelby shear components; failing that, a demonstration that the same constant appears for a different voxel aspect ratio or a different lattice, which no fit could be expected to survive.

9 Reproduction

Every number in this document is produced by scripts in the repository, run under the project’s environment:

- §4–§6, the law and its validation:
`measure_shear_renormalisation.py`
- §2, the prerequisites:
`gate_lateral_sum_invariance.py`, `measure_representation_convergence.py`
- §3, the refutations:
`measure_residual_origin.py`, `measure_lattice_depolarisation.py`
- the exact lattice sum underpinning all of it:
`gate_lattice_kupradze.py`, `gate_lattice_origin.py`, `gate_ewald_kernel_wiring.py`